MILITARY SPECIFICATION

TANK, SPRAY, CHEMICAL AGENT, VX, TMU-28/B; FILLING OF

1. SCOPE

1.1 This specification covers the filing of one type of spray tank with chemical agent VX for use with the tank system in accordance with MIL-S-9822.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-D-3464 — Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.

MIL-S-9822 — System, TMU-28/B, Spray Tank.

MIL-L-25567 — Leak Test Compound, Oxygen Systems.

MIL-D-51083 — Detector Ticket, Anticholinesterase Agent.

MIL-C-51105 — Chemical Agent, VX.

MIL-S-51189 — Substrate Solution and Dispenser.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-1167 — Ammunition Data Card.

MS 20003 — Indicator, Humidity, Card, Three Spot, Impregnated Areas (Cobaltous Chloride).

MS 20995 — Wire, Lock, Carbon Steel, Zinc Coated.

MS 51938 — Seal, Lead and Wire Assembly.

DRAWINGS

AIR FORCE

64J11294 — Tank, Spray, Chemical Agent, VX, TMU-28/B (Data List).

(All drawings referenced in this specification are listed on Dwg 64J11294.)

FSC 1040

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PUBLICATIONS

DEPARTMENT OF THE AIR FORCE
Headquarters, US Air Force—Washington

T.O. 11C15–3–
2–7, Vol. II — Operation and Maintenance Instructions, Shipping and Storage Container, Model CNU–77/E23.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain prior to their use that all materials and components furnished by the supplier have met their respective specifications.

3.2 Empty tank. The tank shall not leak when tested as specified in 4.4.1 (see 4.3.2.1). The tank shall be disassembled and prepared for packing as specified in T.O. 11C15–3–2–7, volume II.

3.3 Filling, closing and marking. The tank shall be filled with 1,398 ± 12 pounds of Government furnished agent conforming to MIL–C–51105 (see 6.2) (166 ± 1.43 gallons at a standard 70°F) and closed and marked in accordance with Drawing 64J11294 (see 4.4.5). When decontaminating fluids are employed, the tank shall be protectively enclosed to prevent contamination or wetting of the tank. Access must be allowed at the filling boss. Prior to or at the hold area, the tank shall be held in such a position that the filling closure (inside the tank) remains completely wetted during leak surveillance. The surveillance period shall be 16 hours minimum unless otherwise specified by the contracting officer. Any protective enclosure shall be removed for reuse or disposal prior to testing in 4.4.2.

3.4 Agent contamination. The simulated tank and the spray tank shall be free from agent contamination when tested as specified in 4.4.2. Immediately after filling and again after storage (see 3.3), the accessible exposed seams of the tank, a random sampling of the tank surface and the welded closure shall be tested for contamination.

3.5 Helium leakage. The welded tank closure shall not leak helium at a rate greater than 1×10^{-6} atmospheric cubic centimeters per second when tested as specified in 4.4.3.

3.6 Container leakage. The container (CNU–77/E23) shall not leak when tested as specified in 4.4.4.

3.7 Serial number. The filled tank shall be marked with a consecutively assigned serial number.

3.8 Workmanship. The spray tank shall be free from visible damage such as cracks, dents or deformation that could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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PEEL ADHESIVE HERE CLEAN WITH SOLVENT

4.3 Inspection provisions.

4.3.1 *Examination and nondestructive tests.* Each tank and container shall be examined in accordance with the list of defects. The acceptance number is 0. Tests 4.4.2 and 4.4.3 will be observed by the Government at the Government's discretion.

4.3.2 *List of defects.*

4.3.2.1 *Tank, spray, chemical agent, TMU–28/B (Dwg 64J11294).*

Categories	*Defects*	*Acceptance standards*
1	Leakage	4.4.1
2	Boss or cap incorrect	(63C57071 and 63C57072)
3	Damage (3.8)	
4	Contamination	4.4.2
5	Helium leakage	4.4.3
6	Marking missing or incorrect	
7	Marking illegible	
8	Test pressure exceeded	4.4.1
9	Serial number missing or incorrect	
10	Component missing or incorrect	
11	Weight incorrect	4.4.5

4.3.2.2. *Preparation for delivery (section 5).*

Categories	*Defects*	*Acceptance standards*
1	Torque of bolts improper	4.4.4
2	Container sealed incorrectly	
3	Component missing or incorrect	
4	Damage (cracks, deformation)	
5	Spare parts packaging incorrect	
6	Desiccant missing or incorrect	
7	Humidity indicator missing or incorrect	
8	Marking missing, illegible or incorrect	
9	Contamination	4.4.2 and 4.4.2.6
10	Lock wire missing or incorrect	
11	Lead seal missing or incorrect	
12	Canister missing or incorrect	
13	Leakage	4.4.4

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4.4. Tests.

4.4.1 *Empty tank.* The empty tank shall be pressurized with dry nitrogen at 8 ± 1 pounds per square inch gage (psig) and shall not leak for 15 ± ½ minutes. Prior to pressurizing a fired cutter housing shall be installed against the outlet diaphragm (Dwg 64D11266) to limit deformation of diaphragm. A pressure of 9 psig shall not be exceeded. If a continuous gage pressure drop is indicated, the tank seams shall be checked for leaks with a liquid leak detector conforming to the performance requirements of MIL–L–25567. A dimensional and surface check shall be made of the press fit surface of the filling boss (Dwg 63C57071) and closure cap (Dwg 63C57072).

4.4.2 *Agent contamination.* The supplier shall submit to the Government procedures for insuring that tanks are properly filled and tanks and containers are not contaminated during process and for decontaminating any contaminated tanks and containers including methods of controlling process. The supplier shall obtain written approval from the Government prior to instituting the process. The supplier may accomplish the minimum work necessary to establish changed procedure without prior Government approval. Subsequently, however, any change to the approved procedure must be approved by the Government in writing before inclusion into the procedure.

4.4.2.1 *Test item.* A simulated tank or protective enclosure as indicated in 3.3 shall be contaminated with liquid agent VX and the spots marked. The test item shall be processed through the decontamination section of the filling line and then tested for agent contamination (see 4.4.2.2).

4.4.2.2 *Spray tank.* The filled spray tank shall be tested for agent contamination as follows: One hundred milliliters of buffer solution (4.4.2.5) shall be poured into a suitable container. A sterile cotton swab, attached to an applicator, shall be dipped in the buffer solution and wiped along the tank seams and other test areas. The cotton swab shall then be replaced in the buffer solution. Stated time intervals for functions below are nominal. The ticket (4.4.2.3) shall be removed from the envelope and immersed in the buffer solution so the sensitive area is below the surface of the buffer solution and held immersed for 15 seconds. The ticket shall then be removed and placed in the envelope where it shall be kneaded for 15 seconds. The ticket shall be removed from the envelope and two drops of substrate solution (4.4.2.4) shall be added to the ticket. The ticket shall be replaced in the envelope and kneaded. If no agent is present, the color of the ticket will turn blue in 60 seconds. If the color of the ticket is not blue, the ticket shall be placed in the envelope and kneaded for an additional 30 seconds. If the color is not blue on the second trial, the test area is considered contaminated. The preceding shall be repeated until all required test areas have been checked for contamination. Evidence of contamination shall be cause for rejection.

4.4.2.3 *Detector ticket.* The ticket shall be in accordance with MIL–D–51083.

4.4.2.4 *Substrate solution.* The substrate solution shall be in accordance with MIL–S–51189. (It is not necessary to use the dispenser or to assemble the solution into the dispenser as specified in MIL–S–51189 for this application.)

4.4.2.5 *Buffer solution.* The buffer solution shall be made up by dissolving 6.07 grams of tris (hydroxymethyl) aminomethane in 250 milliliters (ml) of distilled water; 287 ml of 0.1 normal hydrochloric acid shall be added and the mixture diluted to 1 liter with distilled water. The pH shall be 8.0 ± 0.1.

4.4.2.6 *Toxic agent detection paint.* If Government furnished tanks and containers are coated with a gray detection paint, toxic agent liquid will discolor the gray paint to a bluish-green or a greenish-black. The paint would be applied to the exterior surfaces of

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the tank and as a band at the bottom (Internal) of the shipping container. Precautions must be taken to eliminate liquids of a basic nature to avoid triggering the detection paint. Evidence of contamination shall be cause for rejection.

4.4.3 *Helium leakage.* Prior to closing the tank filling boss (Dwg 63C57071) with the filler cap (Dwg 63C57072) the following conditions shall be met to assure adequate helium leak testing: The void above the agent during the filling cycle shall be cyclically evacuated and pressurized, using helium, between limiting pressures of 12 ± 0.5 and 15 ± 0.5 pounds per square inch absolute (psia) to achieve a satisfactory partial pressure of helium for final closure. Void pressure shall be brought to 0.5 ± 0.5 psig prior to closing. After the closure the weid area shall be sealed off by a test head manifolded to a vacuum system and a mass spectrometer type helium detector. The test head shall be evacuated to and maintained at a pressure lower than 20 microns of mercury during test. The detection system is calibrated by using the standard leak which is built into the system to respond to a leak of 1×10^{-6} atmospheric cubic centimeters per second of helium at the test head.

4.4.4 *Container leakage.* After 5.1 is completed using a torque of 750 to 800 inch pounds on the 60 container closure bolts, the sampling port plug at either end of the container shall be removed and the air relief valves blocked off using an appropriate seal (Duc seal) or a rubber gasketed metal plate. An air pressure of 4.0 ± 0.5 psig shall be introduced through the sampling port. The closure flange and all closure hardware shall be checked for leaks using a liquid leak detector meeting the performance requirements of MIL–L–25567. If leakage occurs, reinstallation and retests shall be performed as required. Necessary repair to Government furnished material shall be approved by the Government.

4.4.5 *Weight.* The net weight of the agent fill shall be verified to be 1398 ± 12 pounds by a weighing device accurate to ± 2.0 pounds.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. Unless otherwise specified in the contract or order, the filled and closed spray tank and components shall be packed in the CNU–77/E23 container as specified in T.O. 11C15–3–2–7, volume II. The technical data including publications if furnished shall be placed in the record receptacle at the right rear section of the container. The explosive cutters will not be packed with the filled tank at the filling facility unless otherwise directed by the contracting officer. Tail cone screws shall be coated with P–9 preservative oil conforming to MIL–P–116 and package screws and slides (Dwg 64C11273) in accordance with method 1 of MIL–P–116. The part numbers shall be stamped in ink on each package. The umbilical cable (Dwg 64C11291) shall be packaged in a polyethylene bag and the bag and other loose parts shall be stored in the two covered storage compartments inside container. The cap assembly (Dwg 63C57073) when removed shall be retained and stored separately for Government disposition. The suspension lugs (1389540) shall be removed from the hard back assembly and individually packaged in accordance with method 1A–8 of MIL–P–116 and stored in accessory compartment. Free movement shall be restricted by use of appropriate dunnage.

5.1.1 *Lock wiring (safety).* Tank fittings (Dwg 65C9042) shall be wired to the yoke assembly with wire per MS 20995. The explosive container assembly (Dwg 65D9019) and the record receptacle bolts shall be wired shut. If data is enclosed wiring shall be as per 5.1.2. The humidity indicator conforming to MS 20003 shall be installed inside the container, immediately behind the humidity indicator viewing disc.

5.1.2 *Lead and wire seal.* The filter and valve cover assembly (Dwg 65E9024) shall be lead and wire-sealed (MC 51938–1). The sampling plugs shall be wired and sealed to plug bosses.

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5.1.3 *Desiccant.* Three hundred and twenty units conforming to MIL–D–3464 shall be added to the desiccant basket of the container and wired shut (see 5.1.1).

5.1.4 *M11 canister.* The canister shall be installed in accordance with T.O. 11C15–3–2–7 Volume II (Dwg 65E9024) and prepare for functioning.

5.1.5. *Tail fins.* The packaged tail fins as received shall be secured inside the CNU–77/E23 container as directed by the contracting officer.

5.2 **Marking.** In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Drawing 65E9008 and MIL–STD–129. Data card shall be in accordance with MIL–STD–1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Material, equipment and facilities to be Government furnished.

6.2 Caution. Due to the hazards of filling munitions with VX, suppliers are advised to exercise all necessary precautions to protect personnel from injury during filling and handling. Care should be taken to prevent wetting the closure opening with agent before welding. A closure opening wetted with agent could result in contamination and inadequate welding making void the results of the helium leak test.

6.3 Contamination. Due to the hazardous nature of VX agent and in order to assure satisfactory test (4.4.2) the following precautions should be taken:

(a) Government approved protective gloves should be worn during test.

(b) If the ticket is applied to the surface, the rubbing should not be such as to disintegrate the ticket. Fresh rubber finger protectors should be used over the gloves for each test to avoid contamination.

6.4 Welding closure. The following recommended weld procedure was found to be satisfactory for closure of the filled tank:

(1) Welding current—D.C. straight polarity.

(2) High frequency—automatic (for arc initiation only).

(3) Pre-purge—Argon 4 seconds.

(4) Cover gas—Argon at 15 cubic feet per hour.

(5) Up-slope—From 10 to 62 amperes in 2 seconds.

(6) Welding—62 amperes for 18.5 seconds.

(7) Down-slope—From 62 amperes to 10.5 amperes in 10 seconds.

(8) Torch speed—1 revolution per 17 seconds.

(9) Cup size—6 or 7.

(10) Tungsten rod—1/16 inch.

6.5 Component specification. The following specification is needed to complete the system:

MIL–S–9822—System, TMU–28/B Spray Tank.

Custodian:

Army—MU

Preparing activity:

Army—MU(EA)

Project No. 1040–A052

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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